

Algebra 2 - Unit 6 Bellwork: Week 1

Name: _____ Date: _____ Period: _____

Day 1 Bellwork: Prior Knowledge Review

Activating what you already know before today's lesson

1. Simplify each expression:

a) $\sqrt[3]{64} = \underline{\hspace{2cm}}$ b) $\sqrt[3]{-27} = \underline{\hspace{2cm}}$ c) $\sqrt[3]{1} = \underline{\hspace{2cm}}$ d) $\sqrt[3]{0} = \underline{\hspace{2cm}}$

2. Rewrite using fractional exponents: $\sqrt[3]{x} = x^{\underline{\hspace{1cm}}}$ $\sqrt[3]{x^2} = x^{\underline{\hspace{1cm}}}$

3. [SAT] What is the slope of the line passing through (2, 5) and (6, 13)?

- A) $\frac{1}{2}$ B) 2 C) 4 D) 8

4. [SAT] Solve for x : $3x - 7 = 2x + 5$

Day 2 Bellwork: Review - Basic Cube Root Solving

Reviewing yesterday's lesson on solving cube root equations

1. Solve: $\sqrt[3]{x} = 5$

2. Solve: $\sqrt[3]{x+4} = 3$

3. Solve: $\sqrt[3]{2x-1} = \sqrt[3]{x+6}$

4. [SAT] A line passes through the points $(-1, 4)$ and $(3, -8)$. What is the equation of the line in slope-intercept form?

A) $y = -3x + 1$

B) $y = -3x + 7$

C) $y = 3x + 1$

D) $y = 3x + 7$

Day 3 Bellwork: Review - Multi-Step & Exponents

Reviewing yesterday's lesson on multi-step equations and fractional exponents

1. Solve and CHECK your answer: $\sqrt[3]{3x - 5} = 4$

2. Solve: $(x + 7)^{1/3} = -2$

3. Solve: $2\sqrt[3]{x - 1} + 3 = 9$

4. [SAT] Solve the inequality: $2x - 5 < 3x + 1$

A) $x > -6$

B) $x < -6$

C) $x > 6$

D) $x < 6$